

Llama-3.3-70B VIS Benchmark

without aiDAPTIV vs with aiDAPTIV (Cold / Warm)

This report compares serving performance with and without aiDAPTIV under two aiDAPTIV states: Cold (first run) and Warm (steady-state average).

1) Benchmark scope

	Model	Llama-3.3-70B-Instruct
	NP levels	16 and 32
	Input tokens	1K, 2K, 4K, 8K, 16K
	Output tokens	1K and 2K

2) Comparison modes

	without aiDAPTIV	3-run baseline average (Run1 / Run2 / Run3)
	with aiDAPTIV (Cold)	Run1 (first run after enabling aiDAPTIV)
	with aiDAPTIV (Warm)	Average of Run2 and Run3 (steady state)

3) Metrics in this report

	TTFT (ms)	Time from request start to the first token.
	TPOT (ms/token)	Average time to generate each token after the first token.
	Throughput (Aggregate tok/s)	Total server output capacity across all users.
	Throughput (Normalized tok/s/user)	Aggregate throughput divided by NP, representing per-user serving capacity.

4) Reading guide

- ✓ TTFT and TPOT: lower is better.
- ✓ Aggregate throughput: total server output capacity.
- ✓ Normalized throughput: aggregate throughput divided by NP, representing per-user serving capacity.
- ✓ Warm-state gains become more visible as input length and concurrency increase.

 This report focuses on NP16 and NP32 serving behavior under short-to-long context workloads.

TTFT

Time to first token (ms). Lower is better. Comparison modes: without aiDAPTIV, Cold, and Warm.

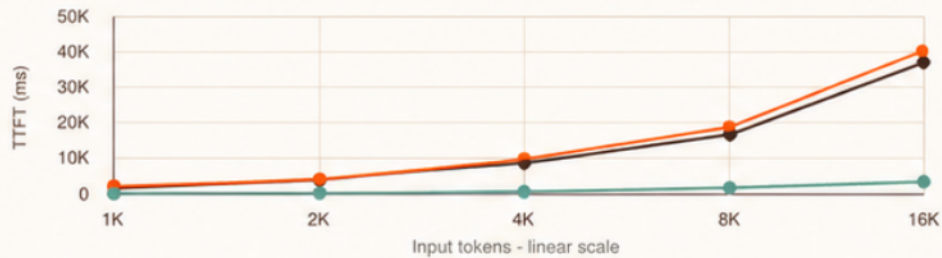
Warm = average of Run2 and Run3; Cold = Run1.

— without aiDAPTIV — Cold (Run1) — Warm (Run2/Run3 avg)

FORMULA $TTFT = t(\text{first token}) - t(\text{request start})$

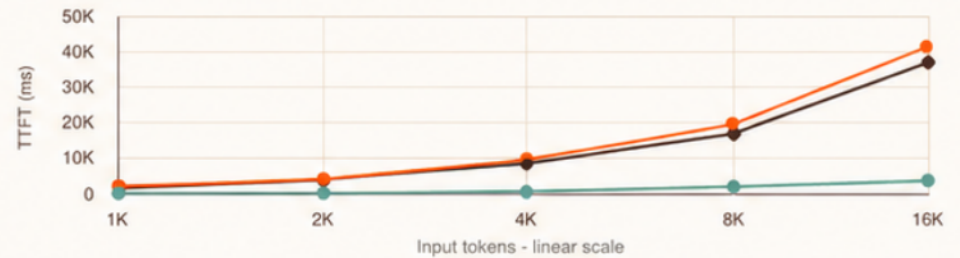
Reduction (%) = $(\text{Without} - \text{Warm}) / \text{Without} \times 100$

NP16 — Output 1K



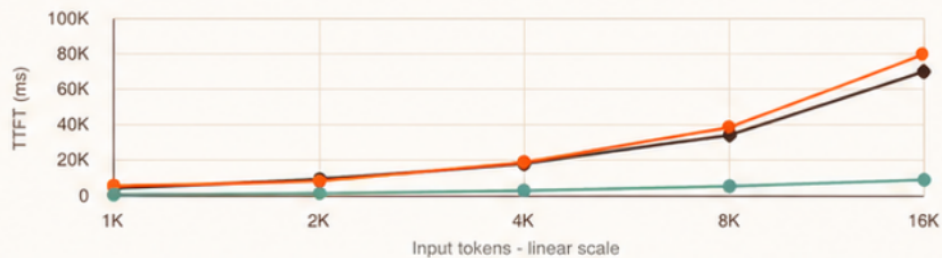
Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	2,254	2,728	157	+93.0% lower
2K	4,932	3,292	319	+93.5% lower
4K	9,373	10,412	749	+92.0% lower
8K	18,780	20,777	1,739	+90.7% lower
16K	37,406	41,238	3,312	+91.1% lower

NP16 — Output 2K



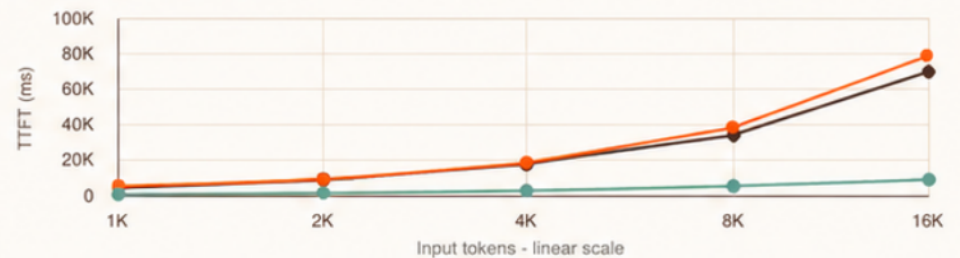
Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	2,554	2,938	153	+94.0% lower
2K	4,857	5,392	303	+93.8% lower
4K	8,995	10,034	433	+95.2% lower
8K	18,667	20,750	1,612	+91.4% lower
16K	37,429	41,162	3,276	+91.3% lower

NP32 — Output 1K



Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	4,668	5,210	293	+93.7% lower
2K	8,825	5,585	604	+93.2% lower
4K	17,331	19,235	1,386	+92.0% lower
8K	35,064	39,011	3,622	+89.7% lower
16K	73,009	79,655	7,632	+89.6% lower

NP32 — Output 2K



Input	Without (ms)	Cold (ms)	Warm (ms)	Improvement vs Without (lower is better)
1K	4,604	5,157	274	+94.1% lower
2K	8,904	9,901	732	+91.8% lower
4K	17,238	19,124	1,259	+92.7% lower
8K	35,305	38,960	3,511	+90.1% lower
16K	72,699	79,712	6,935	+90.5% lower

KEY TAKEAWAY TTFT exceeds the 50% POC goal from NP16 / Input 1K, peaking at 95.2%.

TPOT

Time per output token (ms/token). Lower is better; smaller values *mean faster* decode.

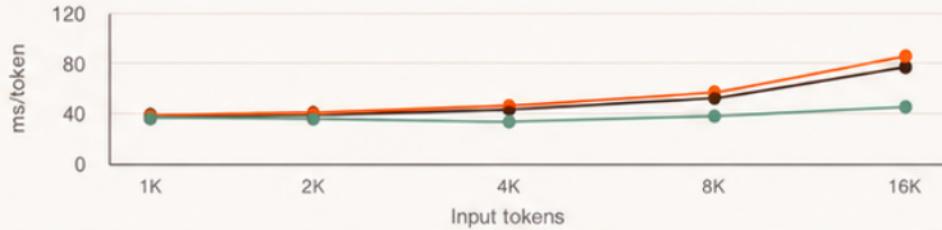
Warm = average of Run2 and Run3, Cold = Run1.

Without (baseline avg) Cold (Run1) Warm (avg of Run2/Run3)

FORMULA $TPOT = (E2E \text{ latency} - TTFT) / (\text{output tokens} - 1)$

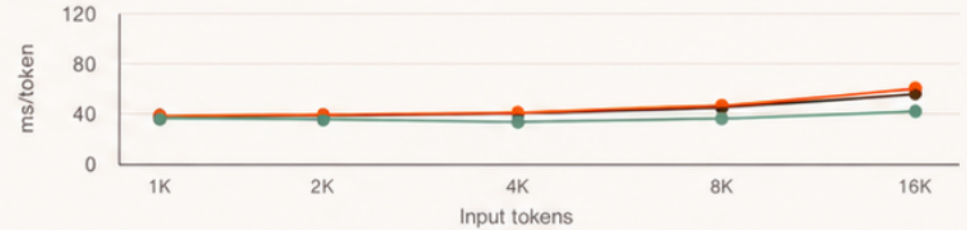
Reduction (%) = $(\text{Without} - \text{Warm}) / \text{Without} \times 100$

NP16 — Output 1K



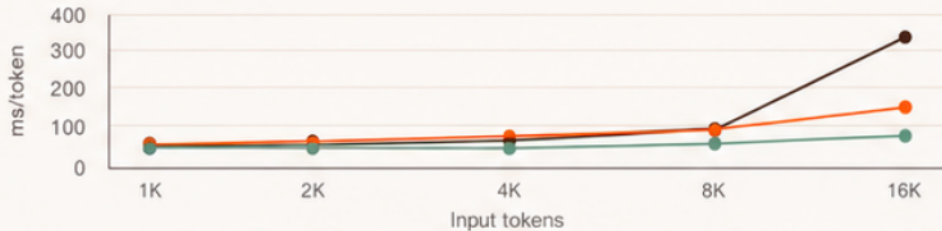
Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	38.74	39.06	38.02	+1.9% lower
2K	41.15	39.92	39.09	+5.0% lower
4K	47.42	48.64	41.92	+11.6% lower
8K	58.18	60.31	45.21	+22.3% lower
16K	87.82	92.34	58.17	+33.8% lower

NP16 — Output 2K



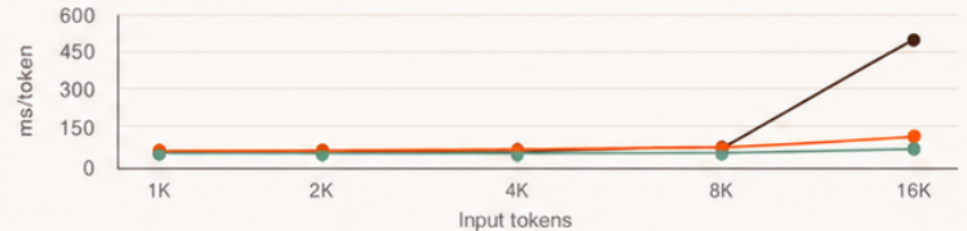
Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	38.43	38.93	38.28	+0.4% lower
2K	42.24	42.72	41.39	+2.0% lower
4K	44.22	45.02	41.91	+5.2% lower
8K	52.47	53.94	46.02	+12.3% lower
16K	67.83	70.52	54.64	+19.4% lower

NP32 — Output 1K



Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	47.88	48.52	45.56	+4.8% lower
2K	53.13	50.45	47.43	+10.7% lower
4K	65.09	67.34	51.92	+20.2% lower
8K	89.47	94.16	60.15	+32.8% lower
16K	332.23	150.38	79.71	+76.0% lower

NP32 — Output 2K



Input	Without (baseline avg)	Cold (Run1)	Warm (avg)	Improvement vs Without (lower is better)
1K	47.36	47.85	46.36	+2.1% lower
2K	51.03	51.87	48.01	+5.9% lower
4K	58.67	60.28	52.49	+10.5% lower
8K	75.00	77.82	60.75	+19.0% lower
16K	491.80	119.33	82.92	+83.1% lower

KEY TAKEAWAY TPOT improves from NP16 / Input 1K and meets the 50% POC goal at NP32 / Input 16K, peaking at 83.1%.

Throughput (Aggregate)

Aggregate output throughput (tok/s). Higher is better; this is total server serving capacity.

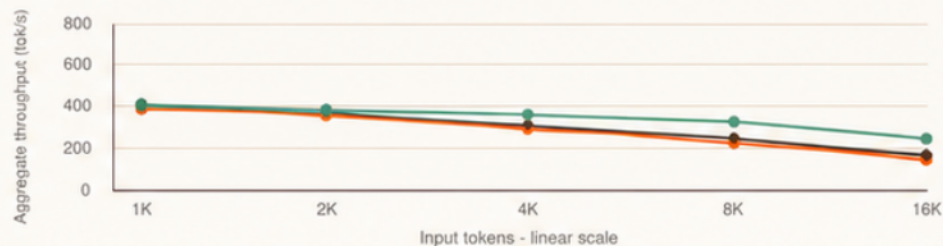
Warm = average of Run2 and Run3, Cold = Run1.

— without aiDAPTIV — Cold (Run1) — Warm (Run2+Run3)

FORMULA Aggregate tok/s = total generated output tokens / benchmark duration

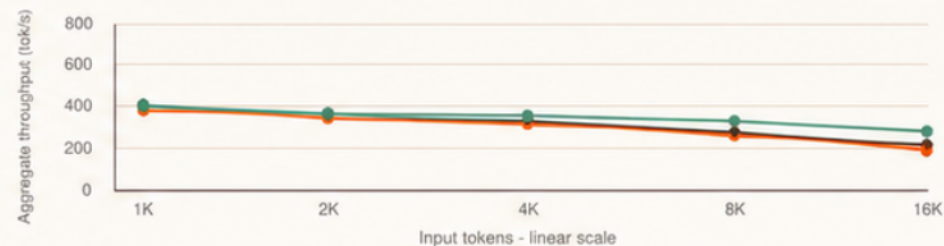
Gain (%) = (Warm / Without - 1) x 100

NP16 — Output 1K



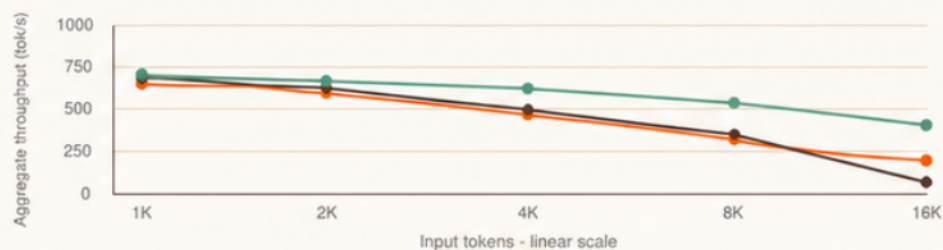
Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	412.96	409.60	420.80	+1.9%
2K	388.80	400.80	409.36	+5.3%
4K	337.44	328.96	381.68	+13.1%
8K	274.99	265.28	353.92	+28.7%
16K	182.19	173.28	275.04	+51.0%

NP16 — Output 2K



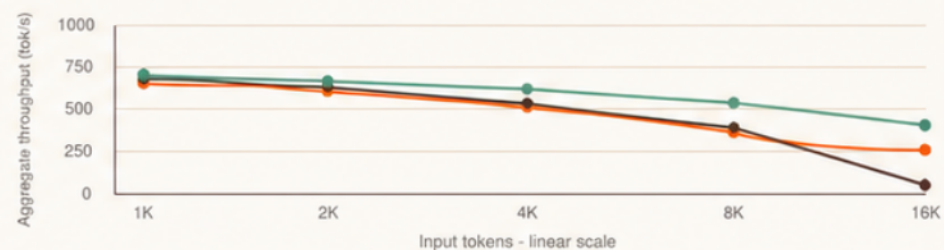
Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	416.32	411.04	418.00	+0.4%
2K	378.83	374.56	386.56	+2.0%
4K	361.81	355.36	381.76	+5.5%
8K	304.96	296.64	347.68	+14.0%
16K	235.89	226.88	292.80	+24.1%

NP32 — Output 1K



Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	668.37	659.52	702.40	+5.1%
2K	602.35	634.24	674.72	+12.0%
4K	491.63	475.20	616.32	+25.4%
8K	357.65	339.84	532.00	+48.8%
16K	96.32	212.80	401.44	+316.8%

NP32 — Output 2K



Input	Without (tok/s)	Cold (tok/s)	Warm (tok/s)	Improvement vs Without (higher is better)
1K	675.63	668.80	690.24	+2.2%
2K	627.09	616.96	666.56	+6.3%
4K	545.39	530.88	609.60	+11.8%
8K	426.67	411.20	526.72	+23.5%
16K	65.07	268.16	385.92	+493.1%

KEY TAKEAWAY Aggregate throughput improves from NP16 / Input 1K; the 50% POC goal is met at Input 16K for NP16 / Output 1K and NP32, peaking at 493.1%.

Throughput (Normalized)

Normalized output throughput (tok/s/user). Higher is better; this is aggregate throughput divided by NP.

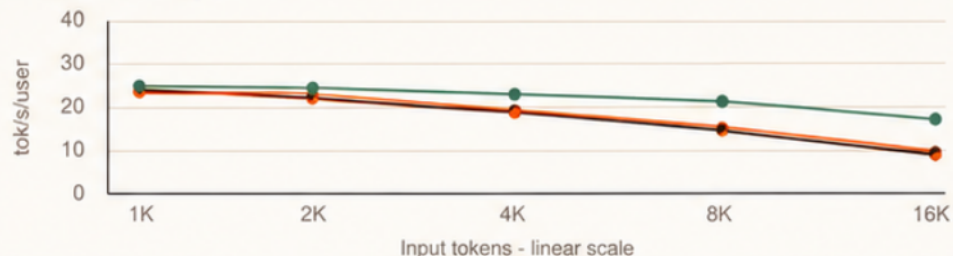
Warm = average of Run2 and Run3; Cold = Run1.

— without aiDAPTIV — Cold (Run1) — Warm (Run2+Run3)

FORMULA Normalized tok/s/user = Aggregate tok/s / NP (concurrent users)

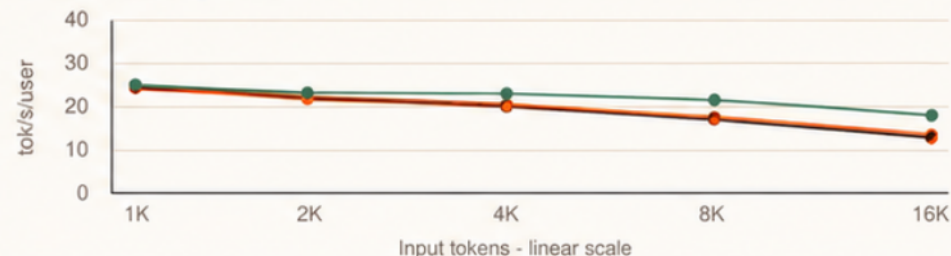
Gain (%) = (Warm / Without - 1) x 100

NP16 — Output 1K



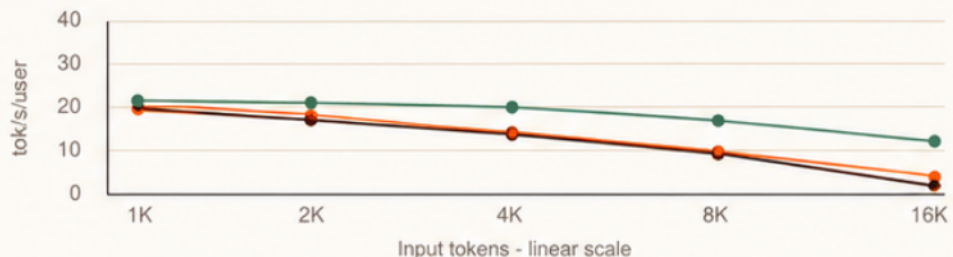
Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	25.81	25.60	26.30	+1.9%
2K	24.30	25.05	25.58	+5.3%
4K	21.09	20.56	23.86	+13.1%
8K	17.19	16.58	22.12	+28.7%
16K	11.39	10.83	17.19	+51.0%

NP16 — Output 2K



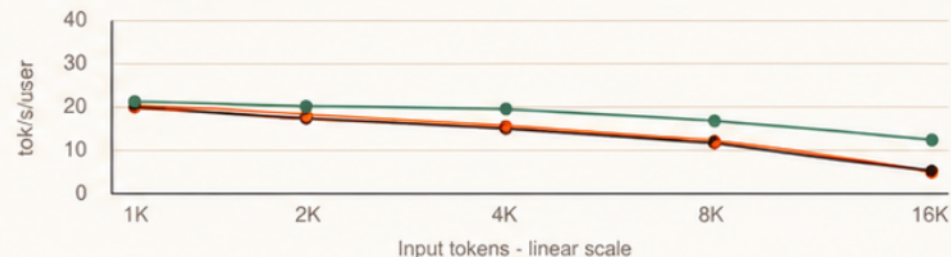
Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	26.02	25.69	26.12	+0.4%
2K	23.68	23.41	24.16	+2.0%
4K	22.61	22.21	23.86	+5.5%
8K	19.06	18.54	21.73	+14.0%
16K	14.74	14.18	18.30	+24.1%

NP32 — Output 1K



Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	20.89	20.61	21.95	+5.1%
2K	18.82	19.82	21.08	+12.0%
4K	15.36	14.85	19.26	+25.4%
8K	11.18	10.62	16.62	+48.8%
16K	3.01	6.65	12.54	+316.8%

NP32 — Output 2K



Input	Without (tok/s/user)	Cold (Run1) (tok/s/user)	Warm (Run2+Run3) (tok/s/user)	Improvement vs Without (higher is better)
1K	21.11	20.90	21.57	+2.2%
2K	19.60	19.28	20.83	+6.3%
4K	17.04	16.59	19.05	+11.8%
8K	13.33	12.85	16.46	+23.5%
16K	2.03	8.38	12.06	+493.1%

KEY TAKEAWAY Per-user throughput improves from NP16 / Input 1K; the 50% POC goal is met at Input 16K for NP16 / Output 1K and NP32, peaking at 493.1%.

SUMMARY

aiDAPTIV impact on latency, decode efficiency, and throughput

Llama-3.3-70B VIS benchmark, NP16 and NP32; without aiDAPTIV vs Cold vs Warm.

1

TTFT (Time to First Token)

Warm aiDAPTIV reduces TTFT by ~89.6%-95.2% across all tested settings.

NP16

Output 1K

90.7%-93.5%

Output 2K

91.3%~~95.2%~~

NP32

Output 1K

89.6%-93.7%

Output 2K

90.1%-94.1%

2

TPOT (Time per Output Token)

NP32 / 16K long-context decode efficiency.

16K input | Output 1K

332.23 -> 79.71

-76.0%

16K input | Output 2K

491.80 -> 82.92

~~-83.1%~~

3

Aggregate Throughput (tok/s)

NP32 / 16K: warm-state gains grow sharply with concurrency.

16K input | Output 1K

96.32 -> 401.44

+316.8%

16K input | Output 2K

65.07 -> 385.92

~~+493.1%~~

4

Normalized Throughput (tok/s/user)

Per-user serving capacity mirrors aggregate scaling gains.

NP16 / Out1K

11.39 -> 17.19

+51.0%

NP16 / Out2K

14.74 -> 18.30

+24.1%

NP32 / Out1K

3.01 -> 12.54

~~+316.8%~~

NP32 / Out2K

2.03 -> 12.06

~~+493.1%~~



KEY
TAKEAWAYS

1

TTFT >50% target achieved

95.2% lower
maximum reduction

2

TPOT >50% target achieved

83.1% lower
NP32 / 16K / Out2K

3

Throughput >50% achieved

+493.1%
NP32 / 16K / Out2K